

LEARNING RESOURCE GUIDE

ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional Participant Workbook

Backup, Restore, and Management Plane Operations

1. Learning Objectives

By the end of this lecture, you will be able to:

- Distinguish between full, incremental, and differential backup types and explain their restore implications.
- Explain RTO and RPO and align backup frequency and testing requirements to each.
- Describe why backup data must be geographically distributed and encrypted with separately managed keys.
- Define the management plane and explain why it is the highest-value target in a cloud environment.
- Identify the access controls and monitoring practices required to protect management plane integrity.

2. Introduction

At 11:47 PM on a Tuesday, a senior cloud engineer at a mid-sized financial technology company received an alert: the primary database cluster was returning errors. She logged into the cloud console. The production environment appeared intact. Then she noticed the backup jobs: all scheduled backup jobs had been deleted three hours earlier. An audit log showed the deletion was performed using a service account with management plane access — a service account that had not been reviewed since the previous year and whose credentials had been rotated so infrequently that the exposure window was measured in months. The attacker had gained access to the management plane, silently deleted the backup configuration, and was preparing to deploy ransomware. The production data was intact. The recovery capability had been destroyed.

That incident illustrates two interlocking themes of this lecture. First, backup and restore are operational security controls — they are not passive archives but active capabilities that require testing, monitoring, and protection. Second, the management plane is not just a convenience layer for infrastructure orchestration; it is the most powerful and therefore the most targeted security boundary in a cloud environment. An attacker who gains management plane access can undo every other security control in place.

3. Core Concepts Explained

3.1 Backup Types: Full, Incremental, Differential

The three foundational backup types represent different trade-offs between backup speed, storage consumption, and restore complexity.

A full backup captures the complete state of the data set at a point in time. It is the most comprehensive and the simplest to restore from — a single backup set contains everything needed. It is also the most time-consuming to create and the most storage-intensive. Organisations typically run full backups on a weekly or monthly schedule, not daily.

An incremental backup captures only what has changed since the last backup of any type — whether that was a full backup or a prior incremental. Incrementals are fast and storage-efficient because they capture only the delta since the last backup. However, restoration is complex: to restore a system to a given point in time, the recovery process must apply the last full backup and then every incremental in sequence since that full. The more incrementals that have accumulated, the longer and more procedurally complex restoration becomes. A failure in any incremental in the chain may make restoration to that point impossible.

A differential backup captures everything that has changed since the last full backup, regardless of how many incrementals or differentials have run since then. This makes each differential larger than an equivalent incremental, but restoration requires only two sets: the last full backup and the most recent differential. For cloud environments where storage cost is a consideration but restore speed and simplicity also matter, differentials often represent a practical middle ground.

3.2 Cloud Snapshots

Cloud platforms provide native snapshot capabilities that capture the state of a volume or instance at a point in time. Cloud snapshots are typically stored as incremental data blocks on the provider's storage infrastructure — only blocks that have changed since the previous snapshot are stored, making them fast and space-efficient. Snapshots can be initiated on a schedule, triggered by an automation event, or taken manually before a planned change.

Critically, snapshots are not a substitute for tested restore procedures. A snapshot that exists in a storage account but has never been used in an actual restore exercise is an assumption, not a proven recovery capability. The restore process must be tested against documented RTO and RPO requirements to be considered a functioning control.

3.3 RTO and RPO

Recovery Time Objective (RTO) defines the maximum acceptable time from a failure or incident to the restoration of normal service. It is a business requirement, set by stakeholders based on the operational and financial impact of downtime, not a technical constraint. If an organisation's RTO for a given system is two hours, the restore process must demonstrably execute within two hours — proven through testing, not assumed based on documentation.

Recovery Point Objective (RPO) defines the maximum acceptable data loss, expressed as the maximum time window of data that can be lost without unacceptable business impact. If the RPO for a transaction processing system is four hours, backups must be taken at least every

four hours. If a failure occurs and the most recent backup is five hours old, the organisation will lose more data than its RPO permits.

RTO and RPO together define the requirements that backup strategy, restore procedures, and geographic distribution must satisfy. A practitioner designs backward from these requirements: what backup frequency is needed to satisfy RPO, and what restore infrastructure and procedures are needed to satisfy RTO? Discovering during an actual incident that the restore process takes six hours when the RTO is two hours is a planning failure, not a technology failure.

3.4 Geographic Distribution of Backups

Storing backups in the same region as production systems is the most common backup architecture error in cloud environments. If a regional outage occurs — whether due to natural disaster, provider infrastructure failure, or a regional security incident — the production environment and the backup capability that should enable recovery are both unavailable simultaneously. Best practice requires replicating backups to a separate region or at minimum a separate availability zone, ensuring that a localised event does not eliminate both the production environment and the recovery options.

Many compliance frameworks — including FedRAMP, HIPAA security rule implementation specifications, and PCI DSS — explicitly require geographic separation of backup storage as a control. For organisations subject to these frameworks, geographic distribution is not a best practice recommendation; it is a compliance requirement.

3.5 Backup Encryption

Backup archives contain the same sensitive data as production systems — often more, since backups accumulate historical data over time that may not exist in the production system's current state. Backup archives must be encrypted both in transit, using TLS or equivalent, and at rest using strong encryption. The encryption key management practice for backup archives is a specific security consideration: encryption keys for backup data should be managed separately from the storage containing the backups. If an attacker gains access to the backup storage and the encryption keys are stored in the same system or under the same access credentials, encryption provides no additional protection. Separating key management from backup storage ensures that storage access alone does not yield readable backup data.

3.6 Host vs. Guest OS Backup

In virtualised cloud environments, backup operates at two distinct layers. Host-level backups protect the underlying infrastructure configuration: hypervisor settings, network configuration, and management tools installed at the host level. Guest OS backups protect the applications and data running within virtual machines — the workloads. Both layers require backup coverage, and the CCSP exam distinguishes between them. Host-level configuration can often be reproduced from infrastructure-as-code templates, but data within guest VMs generally requires explicit backup processes configured within or around each VM.

3.7 The Management Plane

The management plane is the control layer of the cloud environment — the layer where infrastructure is configured, provisioned, and orchestrated. Management plane tools include cloud provider native consoles and APIs, infrastructure-as-code platforms such as Terraform, configuration management tools such as Ansible and Chef, and container orchestration control planes such as Kubernetes. Scheduling, orchestration, and automation of infrastructure operations all occur at the management plane.

Because the management plane controls everything in the environment, it is also the highest-value target for attackers and the highest-consequence access to protect. An attacker with management plane access can: delete production backups and disable backup schedules; modify security group rules to open inbound access from the internet; exfiltrate credentials stored in secrets managers; provision unauthorised resources for cryptomining or as attack infrastructure; and alter audit logging configurations to suppress evidence of their activity. Every other security control in the environment — network segmentation, endpoint protection, identity management — can be bypassed or defeated by an attacker with management plane access.

3.8 Management Plane Security Controls

Management plane access must be protected with the highest available authentication assurance. Multi-factor authentication is the minimum standard; hardware security key authentication or certificate-based authentication provides higher assurance for the most privileged operations. Least-privilege access applies with particular force: no management plane identity should have broader permissions than its specific function requires, and no long-lived credentials with broad management plane permissions should exist without active monitoring and rotation schedules.

Privileged access workstations — dedicated, hardened endpoints used exclusively for management plane access — reduce the risk of credential theft from endpoints where general-purpose activity creates exposure. Time-limited just-in-time credentials, which are provisioned for a specific administrative task and automatically expire, further reduce the standing access footprint that represents persistent risk.

Every API call to the cloud provider's control plane is a management plane event, and those events must be captured in comprehensive audit logs, retained for the required period, and monitored for anomalous patterns. The specific concern is not only failed access attempts — which are visible — but authorised access used in anomalous ways: a service account that has never deleted backup jobs suddenly deleting all backup jobs at 8:47 PM, or a developer identity making management API calls from an unrecognised IP address. Separation of duties between those who can modify management plane configurations and those who approve changes provides a human governance layer on top of technical access controls.

4. Comparison: Full, Incremental, and Differential Backup Types

Attribute	Full	Incremental	Differential
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What is captured	All data	Changes since last backup (any type)	Changes since last full
Backup speed	Slowest	Fastest	Moderate
Storage used	Highest	Lowest per run	Moderate
Restore complexity	Lowest (single set)	Highest (full + all incrementals)	Low (full + one differential)
Restore speed	Fastest	Slowest	Moderate
Typical frequency	Weekly or monthly	Daily	Daily or multiple per week
Failure risk in restore	Lowest	Highest (chain failure)	Low

5. Case Study: Capital One Cloud Misconfiguration and Management Plane Exposure

Incident: In 2019, a former cloud service provider employee exploited a misconfigured web application firewall in Capital One's AWS environment. The misconfiguration allowed the attacker to execute a server-side request forgery (SSRF) attack that returned AWS IAM credentials from the EC2 instance metadata service. Those credentials had management plane permissions that included the ability to list and download data from S3 buckets. The attacker used those permissions to exfiltrate over 100 million customer records. The root cause was a management plane identity — an IAM role — with broader permissions than the application required. The principle of least privilege applied to management plane credentials would have limited the S3 permissions to the specific buckets the application legitimately needed, containing the blast radius even if the credential was stolen.

The Capital One breach is a canonical management plane security case study. It demonstrates that management plane credential theft does not require attackers to compromise a human administrator — it can result from a technical vulnerability in an application that has been granted excessive management plane permissions. Least privilege for all management plane identities, human and non-human alike, is the foundational control.

6. Analogy: The Management Plane as a Building's Master Key System

Think of the management plane as a building's master key system. Individual keys open individual rooms — that's regular user access to specific resources. A floor master opens all rooms on one floor — that's a more privileged role. The grandmaster key opens every door in the building, disables alarms, controls the HVAC, and can lock anyone else out. That's management plane access. The grandmaster key requires the most stringent physical security: it is kept in a secure vault, only specific authorised people can check it out, every checkout is logged, and it is never copied or left unattended. Management plane credentials are the grandmaster key of your cloud environment. Treat them accordingly: minimum-

necessary access, comprehensive audit trails, hardware-backed authentication, time-limited issuance, and immediate revocation when no longer needed.

7. Exam Focus

CCSP exam questions on backup, restore, and management plane operations appear across Domain 5. Key themes:

- Backup type restore implications — know which combination of sets is needed to restore for each type. Full: one set. Incremental: full plus all incrementals. Differential: full plus most recent differential.
- RPO drives backup frequency — if RPO is 4 hours, backups must run at least every 4 hours. This is a direct calculation the exam tests.
- RTO must be tested, not assumed — examiners specifically present scenarios where a restore takes longer than the stated RTO, asking what planning failure this represents.
- Geographic backup distribution — backups in the same region as production fail if the region fails. Separate region is the required best practice.
- Backup encryption key separation — encryption keys should not be stored in the same system as the backup data they protect.
- Management plane as highest-value target — the exam emphasises that management plane access enables an attacker to defeat every other control.
- Least privilege for all identities — the exam tests whether candidates apply least privilege to service accounts and non-human identities, not just human administrators.
- Management plane audit logging — every control plane API call must be logged, retained, and monitored for anomalous patterns.
- Common distractor: equating 'backup exists' with 'backup is tested.' An untested restore is an assumption, not a control.

8. Key Terms Glossary

Full backup: A backup that captures the complete data set at a point in time; simplest to restore from, most storage-intensive.

Incremental backup: A backup capturing only changes since the last backup of any type; fastest and smallest per run, most complex to restore.

Differential backup: A backup capturing all changes since the last full backup; requires only the last full and the most recent differential for restoration.

Snapshot: A point-in-time copy of a storage volume or instance, stored as incremental data blocks by cloud providers.

RTO: Recovery Time Objective; maximum acceptable time from failure to restored service, set by business stakeholders.

RPO: Recovery Point Objective; maximum acceptable data loss expressed as a time window.

Geographic backup distribution: Replicating backup data to a separate region or availability zone to ensure recovery capability survives a localised outage.

Management plane: The control layer of a cloud environment where infrastructure is configured, provisioned, and orchestrated via APIs and tooling.

Least privilege: The security principle that every identity — human or non-human — should have only the permissions required for its specific function.

Just-in-time (JIT) credentials: Time-limited management plane credentials provisioned for a specific task and automatically revoked after expiry.

Privileged access workstation (PAW): A hardened, dedicated endpoint used exclusively for high-privilege management operations.

Separation of duties: A control requiring that the authorisation and execution of high-risk actions be performed by different individuals or roles.

9. Quick Revision Checklist

- ☐ I can explain what a full, incremental, and differential backup captures and what is needed to restore from each.
- ☐ I understand how RPO determines required backup frequency — if RPO is X hours, backup interval cannot exceed X hours.
- ☐ I know that RTO must be tested against actual restore procedures, not estimated from documentation.
- ☐ I can explain why geographic separation of backup storage is required for resilience.
- ☐ I understand why encryption keys for backup data must be managed separately from the backup storage.
- ☐ I can define the management plane and list examples of management plane tools.
- ☐ I can explain why the management plane is the highest-value target in a cloud environment.
- ☐ I know the access controls required for management plane protection: MFA, least privilege, PAWs, JIT credentials, audit logging.
- ☐ I can distinguish between host-level and guest OS backup coverage.
- ☐ I understand why a snapshot that has never been tested is not a proven recovery capability.

10. Practice Questions

Scenario-Based Question 1: SCENARIO: A cloud operations team at a healthcare company runs full backups every Sunday night and incremental backups every night Monday through Saturday. On Thursday at 3 PM, a configuration error corrupts the primary application database. The team must restore the database. The RPO requirement is 24 hours. What backup sets are required to restore to the most recent recovery point before the corruption?

- A) The Sunday full backup only.
- B) The Sunday full backup plus the Monday, Tuesday, and Wednesday incremental backups, applied in sequence.
- C) The Wednesday incremental backup only, since it is the most recent.
- D) The Sunday full backup plus Wednesday's incremental backup only.

Answer: B. Correct. To restore to Wednesday night — the most recent recovery point before Thursday's corruption — the team must apply the last full backup (Sunday) and all incrementals in sequence through Wednesday. Skipping any intermediate incremental would produce an inconsistent restored state.

A: Incorrect. The Sunday full backup alone restores the database to Sunday night's state. Using only the full backup ignores three days of incrementals and would produce a recovery point older than the RPO requirement.

C: Incorrect. An incremental backup contains only the delta since the previous backup, not a complete data set. Restoring from an incremental alone, without the preceding full backup and all intermediate incrementals, does not produce a restorable database state.

D: Incorrect. This approach skips the Monday and Tuesday incrementals, which are required links in the restoration chain for an incremental backup strategy. Without them, the Wednesday incremental cannot be correctly applied.

Scenario-Based Question 2: SCENARIO: An attacker gains management plane access to a company's AWS environment using a stolen IAM access key associated with a service account. Within 20 minutes the attacker has deleted all S3 backup buckets, modified security group rules to allow inbound access from an external IP address, created a new IAM user with administrator permissions, and disabled AWS CloudTrail logging. Which single control, if implemented before the attack, would most directly have limited the attacker's ability to delete backup buckets?

- A) Enabling multi-factor authentication on all human user accounts.
- B) Applying least-privilege permissions to the service account, specifically excluding S3 bucket deletion permissions that the service account's function did not require.
- C) Enabling S3 server-side encryption on all backup buckets.

- D) Requiring VPN access for all AWS console logins.

Answer: B. Correct. A service account with only the permissions needed for its function would not have had permission to delete backup buckets. Least privilege applied to non-human identities is the direct control that would have contained this specific action.

A: Incorrect. MFA on human accounts would not have prevented an attack using a stolen service account access key. Service accounts typically use programmatic access keys rather than console login with MFA.

C: Incorrect. Encryption protects data confidentiality. It does not prevent an identity with the s3:DeleteBucket permission from deleting the bucket — the attacker in this scenario is destroying backups, not reading them.

D: Incorrect. VPN requirements for console access would not constrain API-based access using a stolen access key. Programmatic AWS API calls are made directly to AWS endpoints and are not routed through a console that a VPN policy would govern.

Scenario-Based Question 3: SCENARIO: A financial technology company's disaster recovery plan documents a 2-hour RTO and a 1-hour RPO for its core transaction processing system. The backup team runs hourly incremental backups with a full backup every Sunday. During a quarterly DR test, the team discovers that restoring from the Sunday full backup plus six days of incrementals takes 3.5 hours. What is the correct response to this finding?

- A) Update the DR documentation to reflect a 3.5-hour RTO, since the actual restore time is the authoritative measure.
- B) Revise the backup strategy and restore procedures to achieve the stated RTO — for example by switching to differential backups to reduce restoration chain length, adding restore infrastructure, or pre-staging restore targets.
- C) Accept the finding as a known exception and document it in the risk register without remediation, since 3.5 hours is close to 2 hours.
- D) Reduce the backup frequency to weekly full backups only, which would simplify the restore process to a single backup set.

Answer: B. Correct. The DR test has revealed a gap between the documented RTO and the actual restore capability. The team must address the gap by changing either the backup strategy (fewer steps in the restoration chain), the restore infrastructure (faster systems or pre-staged targets), or the restore procedures — and must validate the improvement through a follow-up test.

A: Incorrect. The RTO is a business requirement set by stakeholders based on acceptable impact, not a technical target that adjusts to match observed restore times. The restore process must be improved to meet the stated RTO.

C: Incorrect. The gap between 2 hours and 3.5 hours represents a 75 percent overrun relative to the RTO. Accepting it without remediation leaves a documented compliance gap and exposes the organisation to greater than agreed impact during an actual incident.

D: Incorrect. Moving to weekly full backups only would dramatically worsen the RPO — from 1 hour to up to 7 days — which fails the RPO requirement. Simplifying the restore at the cost of RPO compliance is not an acceptable trade-off.

Question 4: What is the primary security concern when backup encryption keys are stored in the same system as the backup data they protect?

- A) It increases backup restoration time because the system must decrypt data during the restore process.
- B) A compromise of the backup storage system provides both the encrypted data and the decryption keys, rendering the encryption ineffective.
- C) It prevents the backup from being replicated to a secondary region, violating geographic distribution requirements.
- D) It requires the backup system to store two copies of every encrypted file, doubling storage consumption.

Answer: B. Correct. Encryption provides meaningful protection only when access to the encrypted data does not automatically provide access to the keys needed to decrypt it. Co-locating keys and data in the same system eliminates this protection layer.

A: Incorrect. Decryption time is a performance consideration, not a security concern. The security problem is the risk that both the encrypted data and the keys are accessible together.

C: Incorrect. Key co-location does not technically prevent geographic replication. It is a key management security concern, not a replication constraint.

D: Incorrect. Keys and data are separate artifacts; storing a key alongside the backup does not duplicate the backup data.

Question 5: An organisation's RTO for a production order management system is 4 hours and its RPO is 2 hours. Which backup configuration satisfies both requirements?

- A) Full backup every 6 hours; restore infrastructure capable of completing restoration in 3 hours.
- B) Incremental backup every 2 hours; restore infrastructure and procedures demonstrated to complete restoration in under 4 hours.
- C) Daily full backup; restore infrastructure capable of completing restoration in 3 hours.

- D) Incremental backup every 4 hours; restore infrastructure capable of completing restoration in 2 hours.

Answer: B. Correct. Backups every 2 hours satisfies the 2-hour RPO. A demonstrated restore time under 4 hours satisfies the RTO. Both requirements are met.

A: Incorrect. A 6-hour full backup interval exceeds the 2-hour RPO, meaning a failure could result in up to 6 hours of data loss.

C: Incorrect. A daily full backup cycle means the RPO could be up to 24 hours — far exceeding the 2-hour RPO requirement.

D: Incorrect. A 4-hour incremental cycle means data loss could reach 4 hours, which exceeds the 2-hour RPO. The RTO of 2 hours is met, but the RPO is not.

Question 6: Why is geographic separation of backup storage specifically required rather than simply having redundant copies within the same region?

- A) Because inter-region data transfer is encrypted by default, which provides additional data protection.
- B) Because a regional outage — whether from natural disaster, provider infrastructure failure, or a regional security incident — can simultaneously eliminate both the production environment and any backup copies stored in the same region.
- C) Because cloud providers charge lower storage rates for cross-region replication.
- D) Because regulatory frameworks prohibit storing production data and backup data in the same geographic location.

Answer: B. Correct. Geographic separation ensures that a single regional event cannot destroy both the primary environment and the recovery capability. When both are in the same region, a regional failure makes recovery impossible regardless of how many copies exist within that region.

A: Incorrect. Encryption is not the reason geographic separation is required. Many intra-region transfers are also encrypted. The protection provided by geographic separation is availability, not confidentiality.

C: Incorrect. Cross-region replication typically incurs additional data transfer costs compared to intra-region storage. Cost is not the driver; resilience is.

D: Incorrect. While some frameworks do require geographic separation, this is because of the resilience rationale — not an arbitrary regulatory prohibition. The security reason drives the requirement.

Question 7: Which of the following actions can an attacker with unrestricted management plane access NOT directly perform?

- A) Delete all backup configurations and stored backup data.
- B) Modify network security group rules to open inbound access from the internet.
- C) Exfiltrate data stored in encrypted database volumes where the encryption key is managed by a separate HSM with access controls independent of the management plane identity.
- D) Provision unauthorised compute instances for use as cryptomining or attack infrastructure.

Answer: C. Correct. If the encryption key is managed in a separate Hardware Security Module (HSM) with access controls that are independent of the management plane identity, management plane access alone does not provide the ability to decrypt and read encrypted data. This is exactly why key management separation is a critical control.

A: Incorrect. Management plane access typically includes permissions to modify or delete backup infrastructure, as demonstrated in the introductory scenario.

B: Incorrect. Network configuration changes are management plane operations. An attacker with management plane access can modify security group rules.

D: Incorrect. Provisioning compute resources is a management plane operation. An attacker with management plane access can create, modify, and delete infrastructure resources.

Question 8: What distinguishes a differential backup from an incremental backup in terms of what data is captured?

- A) A differential backup captures data that has changed in the last 24 hours; an incremental captures only the most recently written files.
- B) A differential backup captures all changes since the last full backup; an incremental captures only changes since the last backup of any type.
- C) A differential backup captures the complete data set; an incremental captures only changed files.
- D) A differential backup is stored on-site; an incremental backup is stored off-site for geographic distribution.

Answer: B. Correct. This is the precise technical distinction. A differential grows with each run because it always references the last full. An incremental captures only the delta since the last backup — whether full or incremental — staying small but creating a longer restoration chain.

A: Incorrect. Neither definition is correct. Differential and incremental backups are defined relative to backup reference points — the last full, and the last backup of any type, respectively — not by calendar time windows.

C: Incorrect. A full backup captures the complete data set. A differential captures changes since the last full, not the complete data set.

D: Incorrect. The storage location is an independent architectural choice from the backup type. Differential and incremental are defined by what data is captured, not where it is stored.

Question 9: An organisation implements just-in-time (JIT) credentials for management plane access. What is the primary security benefit of this approach?

- A) JIT credentials allow administrators to access any cloud resource without requiring multi-factor authentication during the session.
- B) JIT credentials reduce the standing access footprint by issuing management plane permissions only for a specific task window, automatically revoking access when the window expires.
- C) JIT credentials replicate management plane permissions across all cloud regions simultaneously, improving operational efficiency.
- D) JIT credentials allow management plane changes to be approved retroactively, improving workflow speed for routine operations.

Answer: B. Correct. Standing management plane credentials represent a persistent risk — if stolen, they can be used indefinitely. JIT credentials are valid only for a specific timeframe and purpose; a stolen JIT credential that has expired provides no usable access.

A: Incorrect. JIT credentials do not bypass MFA. They are time-limited access grants; MFA requirements still apply to the initial authentication that requests the JIT credential.

C: Incorrect. JIT credential systems are about access time-limiting, not cross-region permission replication.

D: Incorrect. JIT credentials are typically part of a request-and-approval workflow that occurs before the access window opens, not retroactive approval.

Question 10: A cloud security team is designing the management plane audit logging strategy for a large AWS environment. Which of the following logging configurations is most important for detecting anomalous management plane activity?

- A) Logging all S3 object reads to identify potential data exfiltration.

- B) Enabling AWS CloudTrail for all regions and accounts, capturing all API calls to the AWS control plane, and forwarding logs to a SIEM for correlation and anomaly detection.
- C) Monitoring guest OS authentication logs on all EC2 instances.
- D) Reviewing the AWS Trusted Advisor dashboard weekly for security findings.

Answer: B. Correct. AWS CloudTrail is the authoritative management plane audit log source. Enabling it across all regions and accounts, forwarding to a SIEM, and applying correlation and anomaly detection is the complete management plane audit logging strategy.

A: Incorrect. S3 object-level logging captures data-plane activity — reads and writes to objects. While valuable for data loss prevention monitoring, it is not the primary management plane audit log source.

C: Incorrect. Guest OS authentication logs are important for endpoint security monitoring but operate at the data plane level within individual instances, not the management plane level.

D: Incorrect. Trusted Advisor provides periodic best-practice assessment recommendations. It is not a real-time management plane audit log that captures individual API calls and enables anomaly detection.